Kinematics Part III: Uniform Circular Motion

1 The Radian and Angular Displacement

1.1 Why Radians over Degrees?

In your previous 1D kinematics studies (like $v = \Delta d/\Delta t$), we dealt with linear distances. When an object moves in a circle, we need to track its rotation. However, the 360° system is historically defined and has **no mathematical connection to physical distances**.

To bridge the gap between a rotation angle and an actual physical distance, physicists exclusively use the **radian (rad)**.

Definition of a Radian: A radian is the angle created when the arc length (s) you travel along the edge of the circle is exactly equal to the radius (r) of the circle.

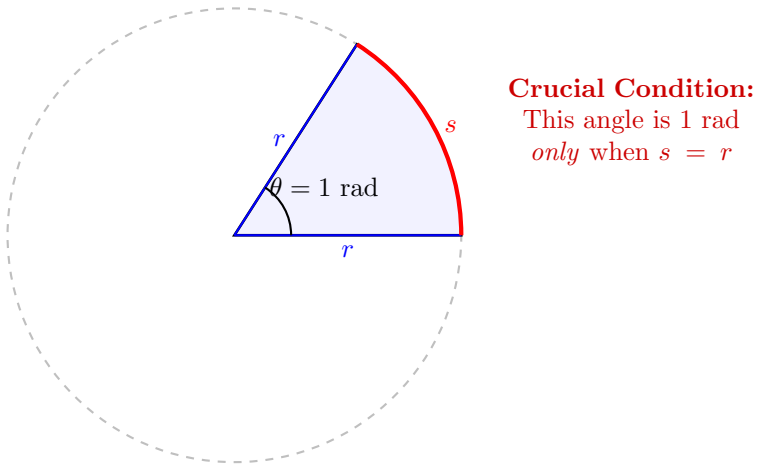


Figure 1: The geometric definition of 1 radian.

1.2 The Bridge Equation ($s = r\theta$)

This definition gives us our first and most important formula linking the angular world to the linear world. If the angle θ is measured in radians, the actual distance traveled along the arc (s) is simply:

$$s = r\theta \tag{1}$$

Converting Degrees to Radians:
We know the circumference of a full circle is $s = 2\pi r$. Plugging this into our equation gives:

$$2\pi r = r\theta \quad \Rightarrow \quad \theta = 2\pi \text{ rad}$$

Thus, $360° = 2\pi$ rad, which gives the standard conversion: $180° = \pi$ rad.

1.3 Angular Displacement ($\Delta\theta$)

In Grade 11, linear displacement ($\Delta\vec{d}$) was defined as a change in position. In rotational kinematics, we use **Angular Displacement ($\Delta\theta$)** to represent the angle an object rotates through.

$$\Delta\theta = \theta_f - \theta_i \tag{2}$$

- Like linear displacement, it is a **vector** (it has direction).
- **Sign Convention:** By standard physics rules, **Counter-Clockwise (CCW)** is defined as positive (+), and **Clockwise (CW)** is negative (−).

2 Angular Velocity and Acceleration

With angular displacement defined, we can express how fast this angle changes.

2.1 Angular Velocity (ω)

Angular velocity (ω, the Greek letter omega) is the rate at which angular displacement changes over time.

$$\omega = \frac{\Delta\theta}{\Delta t} \tag{3}$$

Note: All points on a rigid rotating object (like a spinning merry-go-round) share the exact same ω, no matter how far they are from the center.

2.2 Angular Acceleration (α)

Angular acceleration (α, the Greek letter alpha) is the rate of change of angular velocity:

$$\alpha = \frac{\Delta\omega}{\Delta t} \tag{4}$$

3 The Link Between Linear and Angular (Derivations)

The primary advantage of using radians is that the linear distance s (arc length) traveled by a particle is simply the product of the radius and the angular displacement ($s = r\theta$).

If we observe the motion over a **very short time interval** (Δt), the change in arc length is directly proportional to the change in angle:

$$\Delta s = r(\Delta\theta)$$

By dividing both sides of this equation by the tiny time interval (Δt), we reveal a direct connection between the tangential speed v and angular velocity ω:

$$\frac{\Delta s}{\Delta t} = r\left(\frac{\Delta\theta}{\Delta t}\right) \quad \Rightarrow \quad v = r\omega \tag{5}$$

Crucial Concept: While $\Delta s/\Delta t$ mathematically gives an *average* speed, if we imagine Δt becoming infinitesimally small (approaching zero), it perfectly describes the **instantaneous** values precisely at a single geometric point. This is the foundational idea of Calculus!

Using the exact same algebraic trick over that extremely small Δt, substituting $v = r\omega$, we can connect tangential acceleration a_t and angular acceleration α:

$$\frac{\Delta v}{\Delta t} = r\left(\frac{\Delta\omega}{\Delta t}\right) \quad \Rightarrow \quad a_t = r\alpha \tag{6}$$

These relationships allow us to constantly switch between reading v on a speedometer and calculating the ω of the spinning tires:

Angular Variable	**Linear Counterpart (Translation)**
Angular Displacement: θ	Arc Length (Distance): $s = r\theta$
Angular Velocity: ω	Tangential Speed: $v = r\omega$
Angular Acceleration: α	Tangential Acceleration: $a_t = r\alpha$

3.1 Period (T) and Frequency (f)

Before deriving centripetal acceleration, we must link ω to two macro-scale concepts:

- **Period (T):** The time in seconds (s) to complete exactly **one full revolution** (2π rads).
- **Frequency (f):** The number of full revolutions completed in **one second** (Hz or s^{-1}).

They are reciprocals: $T = 1/f$. Therefore, the angular velocity can be directly calculated if we know how long one lap takes:

$$\omega = \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T} = 2\pi f \tag{7}$$

4 Centripetal Acceleration (a_c) Derivation

Unlike linear motion, an object in uniform circular motion is **always** accelerating, even if its speed v is perfectly constant. Why? Because the *direction* of the velocity vector is constantly changing.

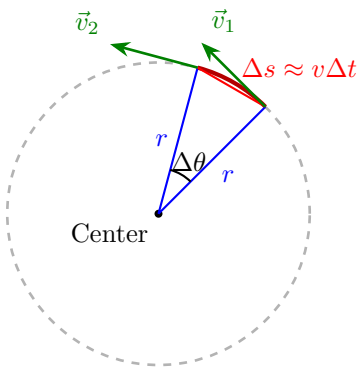
This center-seeking acceleration is called **centripetal acceleration** ($\vec{a}_c$). At any given moment, the velocity vector $\vec{v}$ is tangent to the circle, and the acceleration $\vec{a}_c$ points exactly to the center, meaning $\vec{v} \perp \vec{a}_c$.

4.1 Geometric Proof Using Similar Triangles

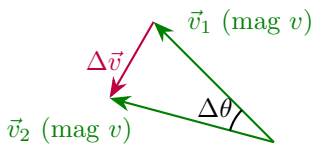
Consider an object moving from a generic point A to point B over a tiny time interval Δt. Its position vector sweeps out an angle $\Delta\theta$, and it travels an arc length roughly equal to a straight chord $\Delta s \approx v\Delta t$.

Meanwhile, the velocity vectors $\vec{v}_1$ and $\vec{v}_2$ (which are both of magnitude v) also rotate by that exact same angle $\Delta\theta$.

1. **The Position Triangle:** Formed by radius r, radius r, and base chord Δs.
2. **The Velocity Triangle:** Formed by the vectors $\vec{v}_1$, $\vec{v}_2$, and the difference vector $\Delta\vec{v}$.
3. Because the central angle $\Delta\theta$ is identical in both, they are **similar isosceles triangles**.



(a) The Position Triangle



(b) The Velocity Triangle ($\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$)

Figure 2: Visualizing the Similar Triangles: Since both sets of vectors sweep through the exact same angle $\Delta\theta$, the physical position triangle is geometrically similar to the velocity vector triangle.

By setting up the ratio of corresponding sides from these similar triangles:

$$\frac{\Delta v}{v} = \frac{\Delta s}{r} \tag{8}$$

Since we know $\Delta s = (v)(\Delta t)$, we can substitute it in:

$$\frac{\Delta v}{v} = \frac{v\Delta t}{r}$$

Divide both sides by Δt to isolate $\frac{\Delta v}{\Delta t}$ (which is exactly the definition of acceleration $\vec{a}_c$):

$$a_c = \frac{\Delta v}{\Delta t} = \frac{v \cdot v}{r}$$

This elegantly gives us the holy grail equation for the *magnitude* of circular motion:

$$a_c = \frac{v^2}{r} \tag{9}$$

What about the direction? Look closely at the Velocity Triangle in Figure (b). The purple vector $\Delta\vec{v}$ (which dictates the direction of our acceleration $\vec{a}_c$) generally points inward. Imagine making the time interval Δt infinitesimally small. As the angle $\Delta\theta$ shrinks to nearly zero, the base of this skinny isosceles triangle ($\Delta\vec{v}$) becomes exactly perpendicular (90°) to the two velocity vectors. Since the velocity is tangent to the circle edge, a vector exactly perpendicular to the tangent must point **directly toward the center** of the circle. This explicit visual proof is exactly why it is named **centripetal** ("center-seeking") acceleration.

4.2 Alternative Forms Using ω and T

By substituting our bridge equation ($v = r\omega$) into the holy grail equation, we get the rotational version:

$$a_c = \frac{(r\omega)^2}{r} = \omega^2 r \tag{10}$$

And by substituting $\omega = 2\pi/T$, we can find the acceleration strictly from the radius and the period:

$$a_c = (\frac{2\pi}{T})^2 r = \frac{4\pi^2 r}{T^2} \tag{11}$$

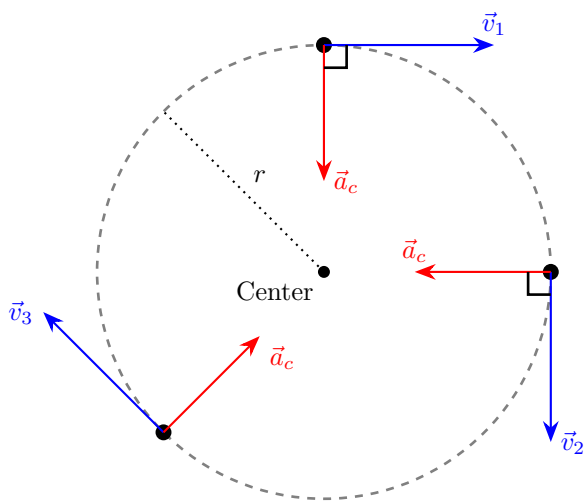


Figure 3: In Uniform Circular Motion, velocity $\vec{v}$ is always tangent, and acceleration $\vec{a}_c$ always points to the center. Thus, $\vec{v} \perp \vec{a}_c$.

5 Total Acceleration (Non-Uniform Circular Motion)

So far, we have largely focused on **Uniform Circular Motion (UCM)**, where the object's speed v remains perfectly constant. In this scenario, the *only* acceleration present is the center-seeking centripetal acceleration ($\vec{a}_c = v^2/r$) responsible for changing the direction.

But what if a car is driving in a circle *and* stepping on the gas pedal at the same time?

In this case, the object is undergoing **Non-Uniform Circular Motion**. We must deal with two independent components of acceleration simultaneously:

- **Centripetal Acceleration ($\vec{a}_c$):** Changes the *direction*. It always points towards the center of the circle.
- **Tangential Acceleration ($\vec{a}_t$):** Changes the *speed* (magnitude of velocity). It is always tangent to the circle (parallel to $\vec{v}$). From our bridge equations, $a_t = r\alpha$.

Since these two vectors are always perfectly perpendicular ($\vec{a}_c \perp \vec{a}_t$), we can find the **Total Acceleration ($\vec{a}_{\mathbf{tot}}$)** using the Pythagorean theorem and simple trigonometry:

$$a_{\text{tot}} = \sqrt{a_c^2 + a_t^2} \tag{12}$$

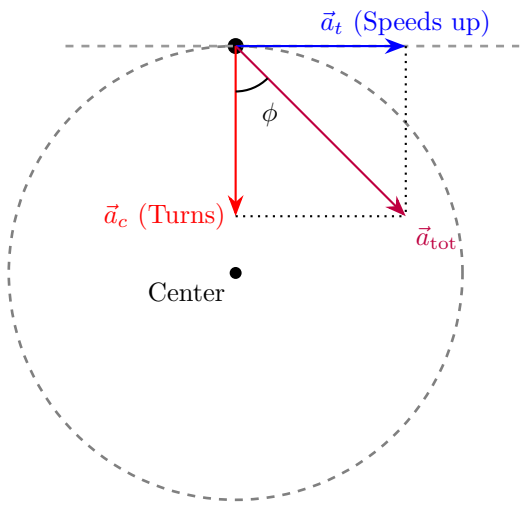


Figure 4: In Non-Uniform Circular Motion, the total acceleration $\vec{a}_{\text{tot}}$ is the vector sum of both the centripetal and tangential accelerations.

The angle ϕ that the total acceleration makes with the radial direction (pointing to the center) can be found using the inverse tangent function:

$$\phi = \tan^{-1}\left(\frac{a_t}{a_c}\right) \tag{13}$$

This concludes the complete kinematic picture of an object moving in a circle, whether its speed is steady or changing!